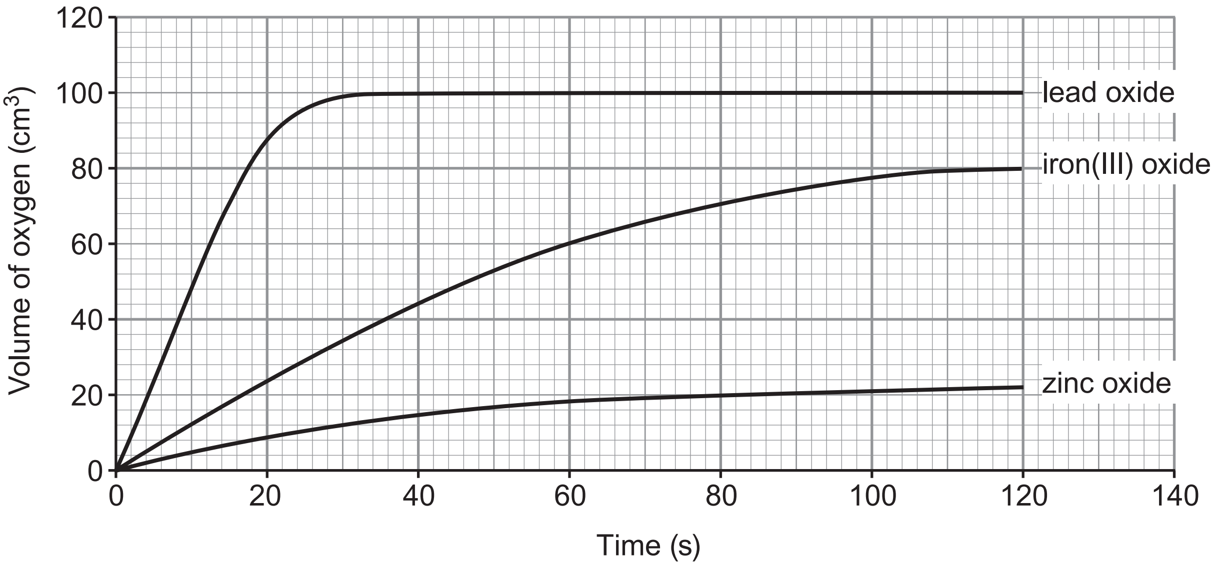
Examiner only

5. Hydrogen peroxide solution decomposes to give water and oxygen gas. Many metal oxides act as catalysts for the reaction.

(a) Use the symbol equation for the reaction to find the formula of hydrogen peroxide. [1]

$$2 \ \ldots\ldots\ldots\ldots \longrightarrow 2H_2O \ + \ O_2$$

(b) Catalysts are used to speed up the reaction. The graphs show the volume of oxygen formed over 120 seconds when 1.0 g of different metal oxides were added to 50 cm³ of hydrogen peroxide solution.



(i) Use the graph to find the time taken to form 44 cm^3 of oxygen using iron(III) oxide as the catalyst. [1]

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(ii) State which metal oxide is the **best** catalyst. Give a reason for your answer. [1]

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(iii) After 120 seconds, the contents of each beaker were washed into a filter paper and funnel. The catalyst left over on each filter paper was dried and weighed.

Tick (✓) the box next to the statement which is correct. [1]

the same mass of all catalysts is left over ☐

more zinc oxide is left over than lead oxide ☐

about 80 % of the iron(III) oxide is left over ☐

no lead oxide is left over ☐

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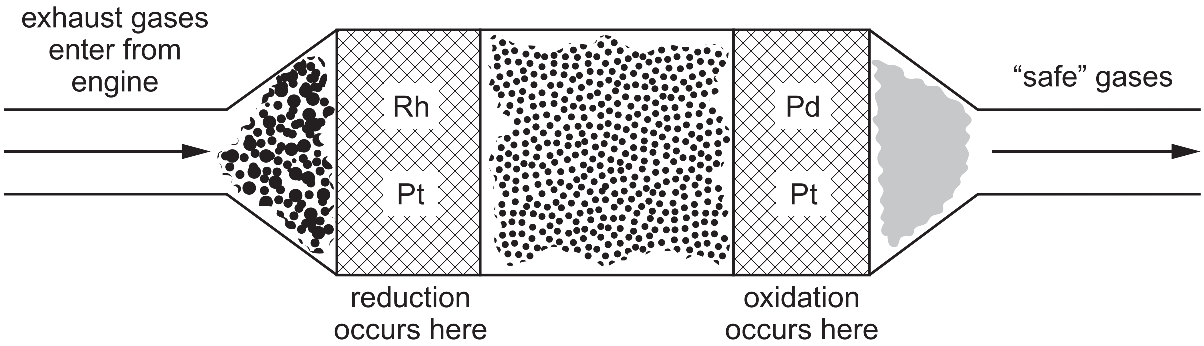
(c) Exhaust gases from car engines contain harmful molecules, for example, carbon monoxide and nitrogen oxides. All cars are fitted with catalytic converters that split up these harmful molecules.

The catalysts are made from platinum (Pt) and palladium (Pd) or rhodium (Rh). A mesh structure is used that exposes the maximum surface area of catalyst to the exhaust gases, while also reducing the amount of catalyst required. Platinum, palladium and rhodium are extremely expensive.

As the exhaust gases from the engine pass over the catalysts, chemical reactions take place on their surfaces. The harmful molecules are broken up and converted into other gases that are “safe” to enter the air. These gases include carbon dioxide, nitrogen, oxygen and water.

Catalytic converter



There are two different types of catalyst, a **reduction catalyst** and an **oxidation catalyst**. The reduction catalyst removes oxygen from nitrogen oxides to make nitrogen and oxygen. The oxidation catalyst adds oxygen to carbon monoxide to make carbon dioxide.

One of the biggest drawbacks of the catalytic converter is that it only works effectively at high temperatures. The average running temperature of a car engine is between 90 and 110 °C. It takes nearly 30 minutes for these temperatures to be reached.

The table opposite shows the percentages of carbon monoxide and nitrogen oxides converted to safe gases by a catalytic converter at different temperatures.

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Temperature (°C)	Carbon monoxide converted (%)	Nitrogen oxides converted (%)
25	16	25
50	19	28
75	26	35
100	60	72
125	91	92
150	93	94
175	95	95
200	97	98

(i) Using your knowledge of particle theory suggest why a catalyst in mesh form works better than a lump of catalyst. [2]

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(ii) Tick (✓) the box that best describes the adverse effect that gases leaving the exhaust would have on the environment. [1]

they have no effect on the environment	☐
they reduce oxygen levels	☐
they deplete the ozone layer	☐
they cause global warming	☐

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(iii) Tick (✓) the box that best describes the conversion of carbon monoxide and nitrogen oxides into "safe" gases at different temperatures. [1]

equal amounts of carbon monoxide and nitrogen oxides are converted at every temperature ☐

more carbon monoxide is converted than nitrogen oxides up to 100 °C ☐

more nitrogen oxides are converted than carbon monoxide up to 100 °C ☐

40 % more nitrogen oxides are converted than carbon monoxide up to 100 °C ☐

(iv) In your opinion how effective are catalytic converters? Explain your answer. [2]

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